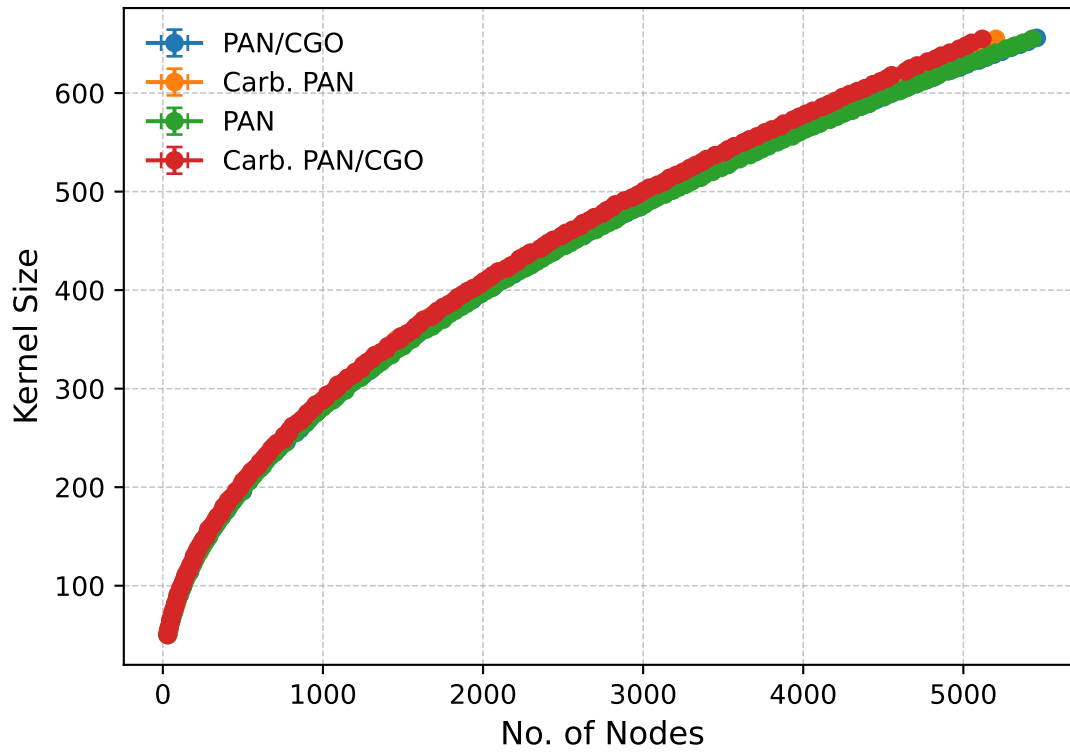


Nodes vs Kernel Size (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Power-law → KS=2.422, p=0.376
 Exponential → KS=inf, p=0.000
 Log-normal → KS=2.074, p=0.512

Carb. PAN:

Power-law → KS=2.431, p=0.372
 Exponential → KS=inf, p=0.000
 Log-normal → KS=2.073, p=0.507

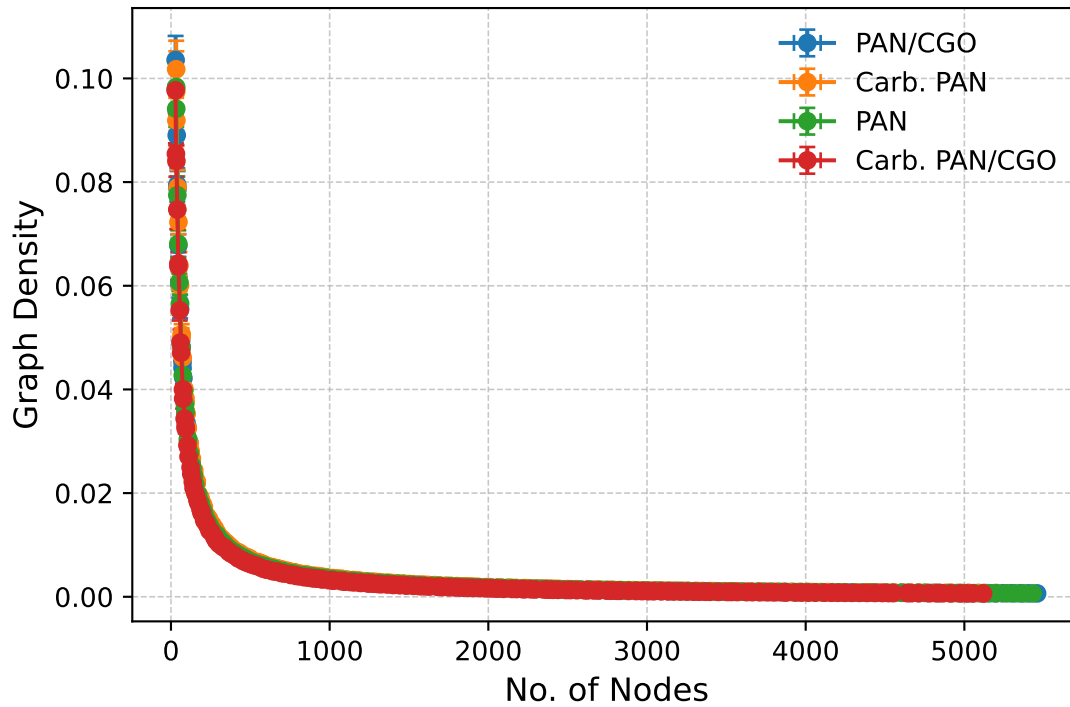
PAN:

Power-law → KS=2.431, p=0.377
 Exponential → KS=inf, p=0.000
 Log-normal → KS=2.073, p=0.510

Carb. PAN/CGO:

Power-law → KS=2.431, p=0.373
 Exponential → KS=inf, p=0.000
 Log-normal → KS=2.073, p=0.507

Nodes vs Graph Density (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Power-law → KS=20.320, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.096, p=0.999

Carb. PAN:

Power-law → KS=19.353, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.092, p=1.000

PAN:

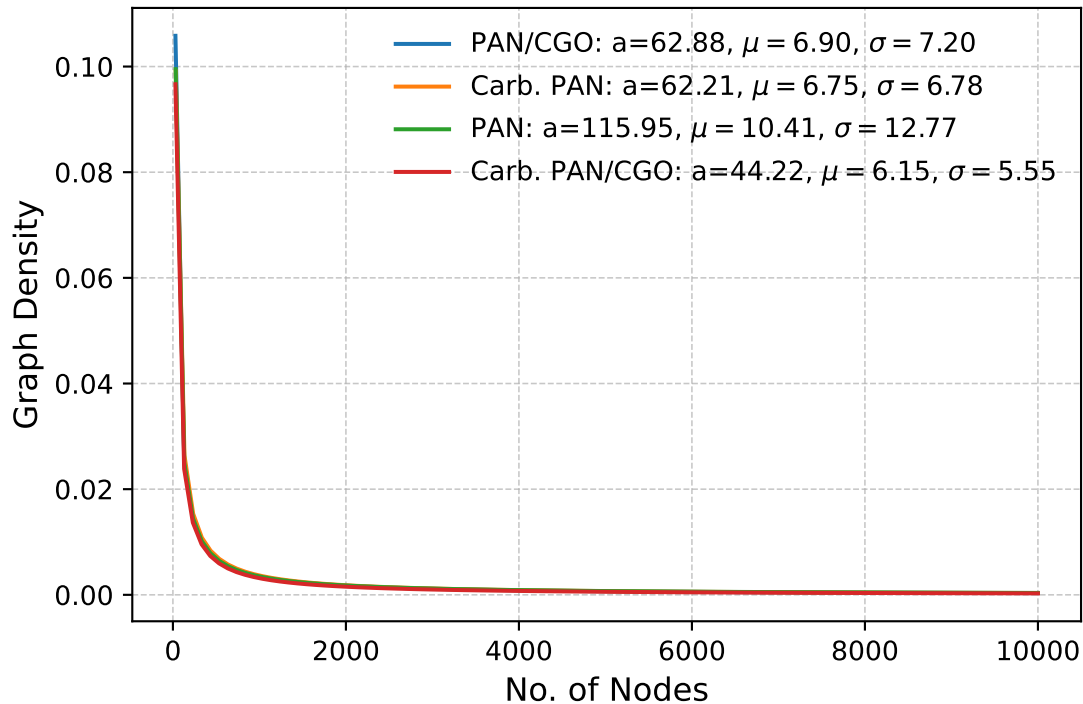
Power-law → KS=19.755, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.086, p=1.000

Carb. PAN/CGO:

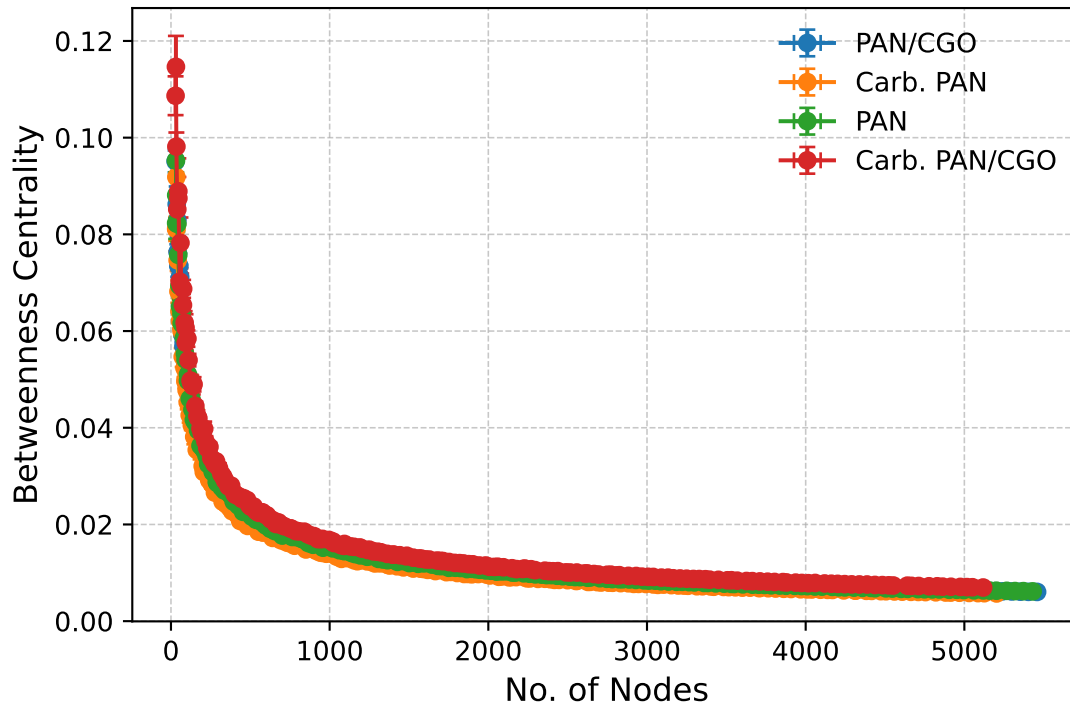
Power-law → KS=19.820, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.095, p=0.999

LogNormal Fit: $y = a \cdot \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$

Nodes vs Graph Density



Nodes vs Betweenness Centrality (Actual Data)



Kolmogorov–Smirnov & P-Values

PAN/CGO:

Power-law → KS=13.883, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.357, p=0.388

Carb. PAN:

Power-law → KS=14.719, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.345, p=0.427

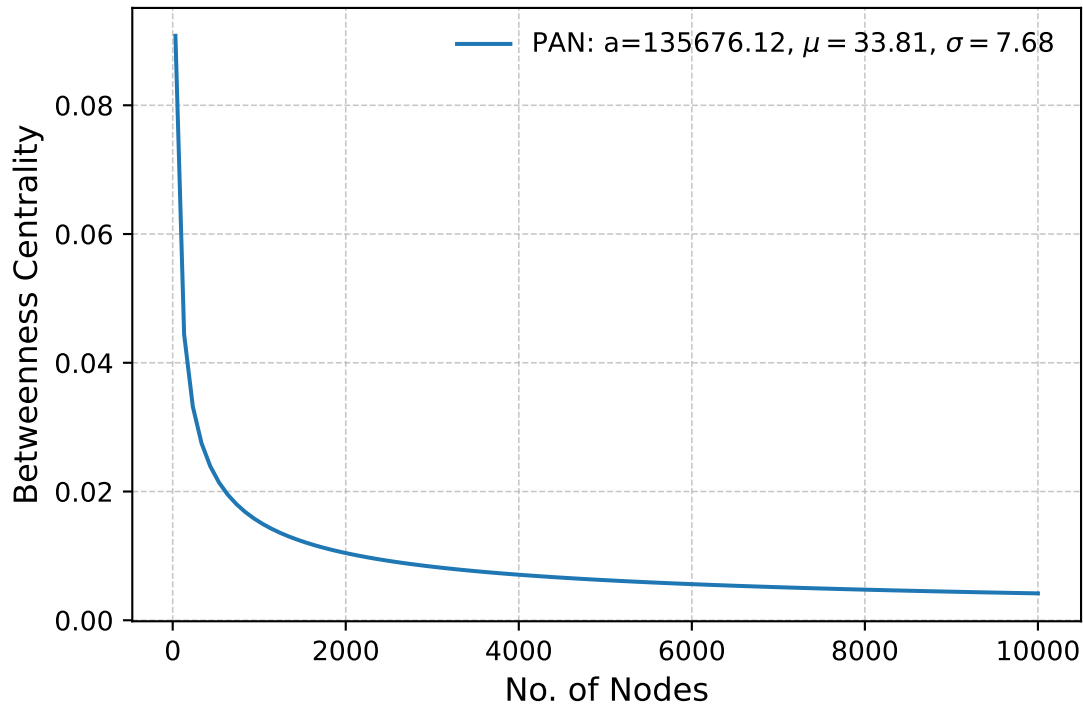
PAN:

Power-law → KS=13.766, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.385, p=0.339

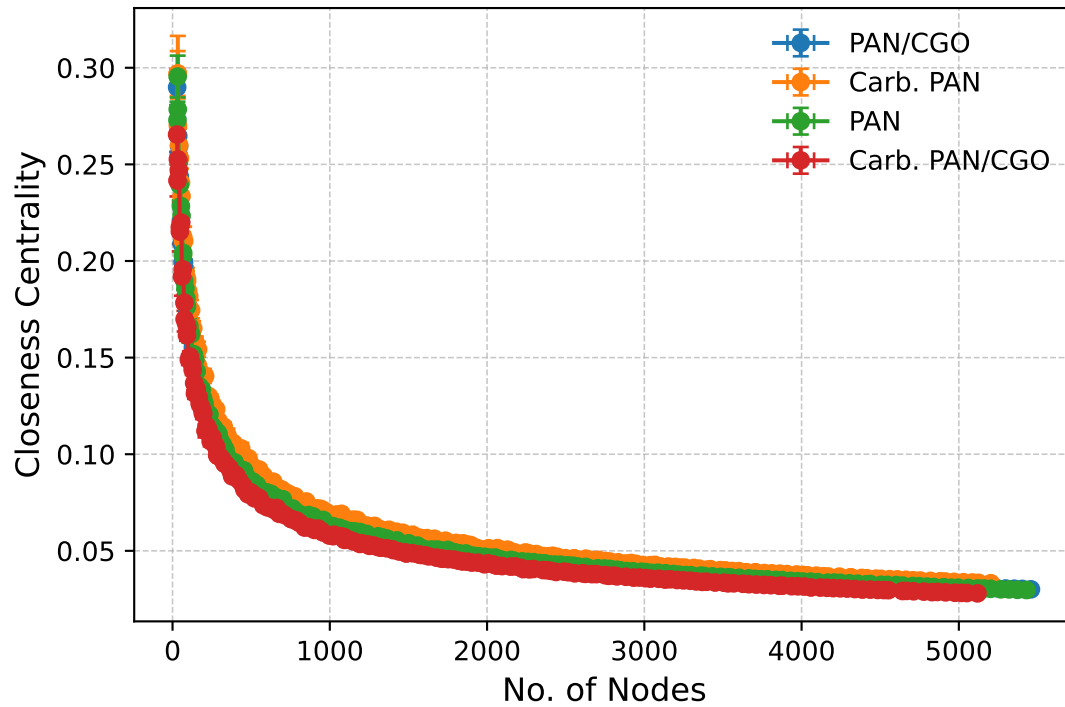
Carb. PAN/CGO:

Power-law → KS=14.880, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.346, p=0.422

LogNormal Fit: $y = a \cdot \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$
 Nodes vs Betweenness Centrality



Nodes vs Closeness Centrality (Actual Data)



Kolmogorov–Smirnov & P-Values

PAN/CGO:

Power-law → KS=12.605, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.472, p=0.180

Carb. PAN:

Power-law → KS=11.940, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.492, p=0.158

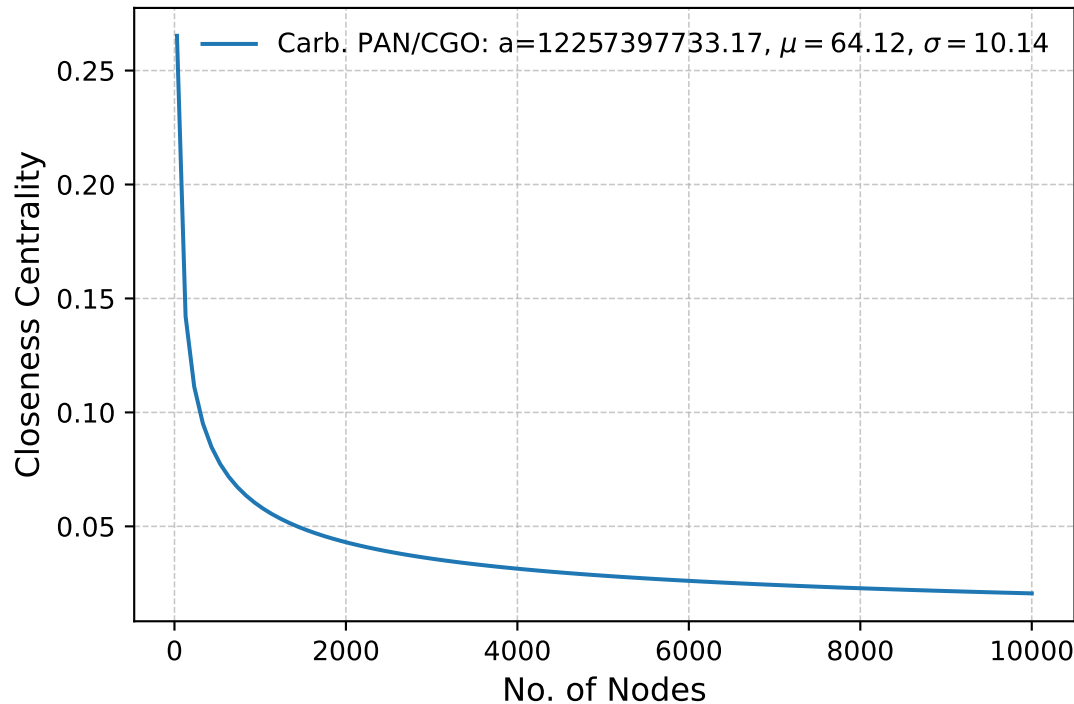
PAN:

Power-law → KS=12.972, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.444, p=0.233

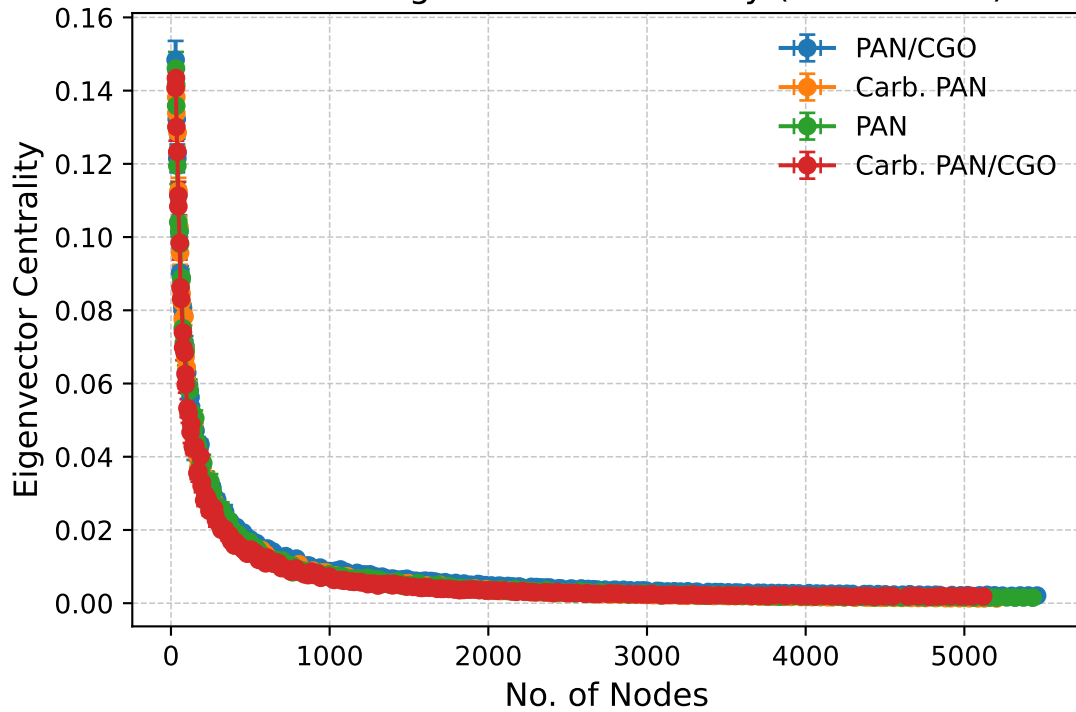
Carb. PAN/CGO:

Power-law → KS=11.834, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.481, p=0.172

LogNormal Fit: $y = a \cdot \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$
 Nodes vs Closeness Centrality



Nodes vs Eigenvector Centrality (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Power-law → KS=17.355, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.185, p=0.900

Carb. PAN:

Power-law → KS=14.848, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.246, p=0.731

PAN:

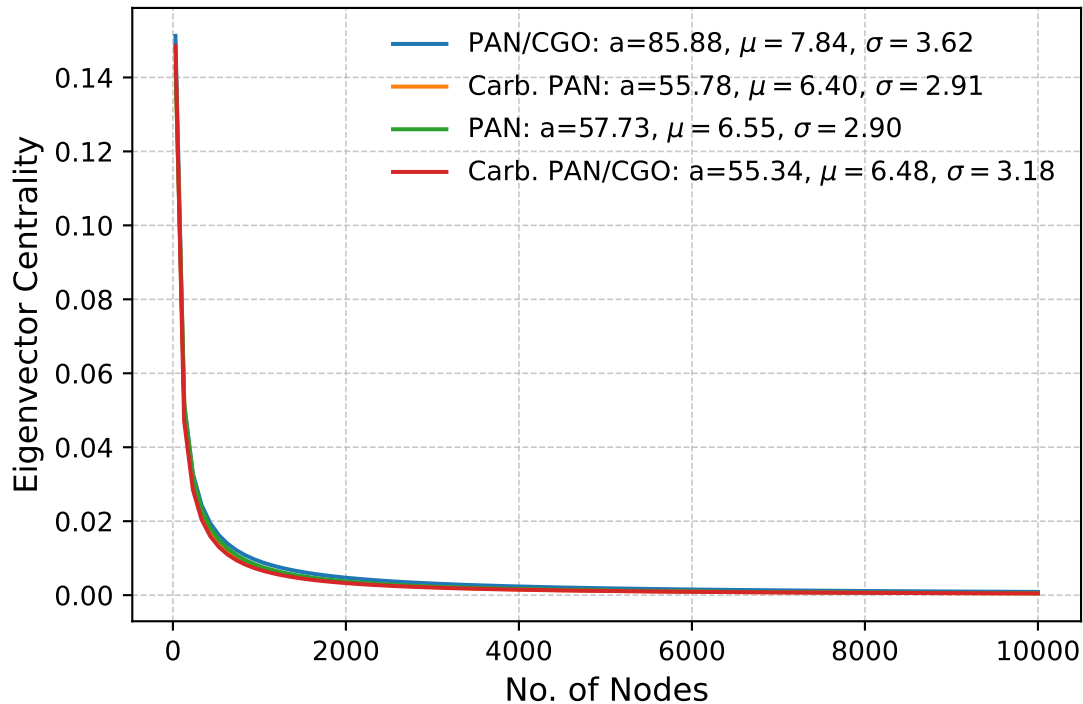
Power-law → KS=13.693, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.414, p=0.330

Carb. PAN/CGO:

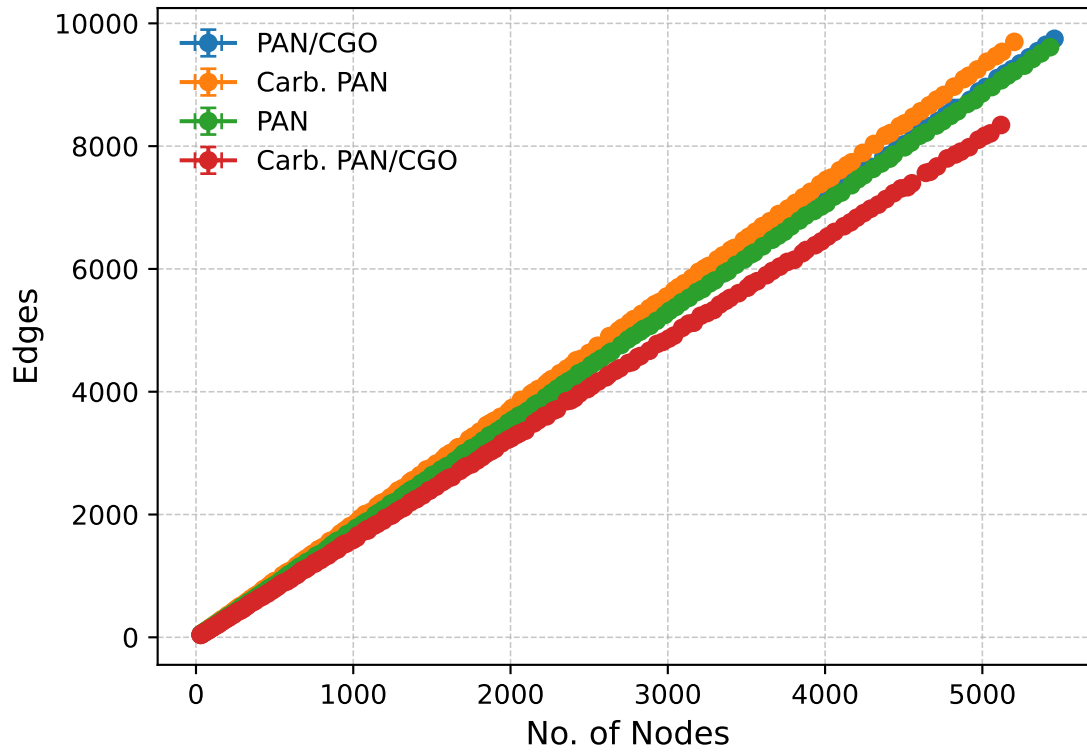
Power-law → KS=16.631, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=0.739, p=0.041

$$\text{LogNormal Fit: } y = a \cdot \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

Nodes vs Eigenvector Centrality



Nodes vs Edges (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Power-law → KS=0.662, p=0.994
 Exponential → KS=inf, p=0.000
 Log-normal → KS=3.495, p=0.523

Carb. PAN:

Power-law → KS=0.755, p=0.964
 Exponential → KS=inf, p=0.000
 Log-normal → KS=3.473, p=0.558

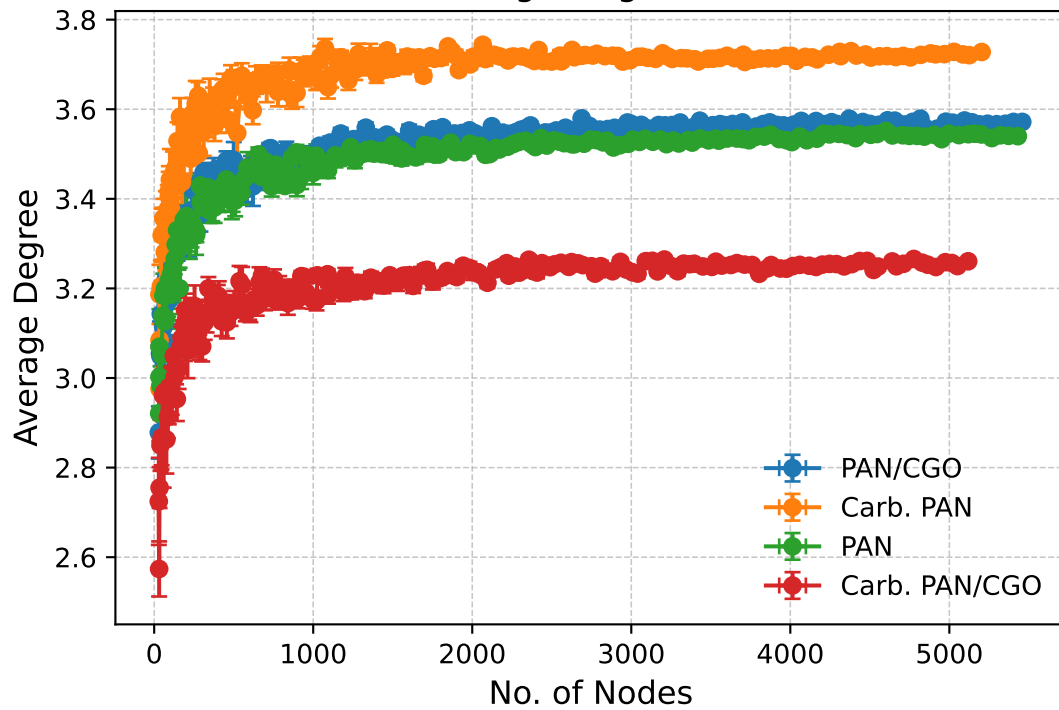
PAN:

Power-law → KS=0.692, p=0.988
 Exponential → KS=inf, p=0.000
 Log-normal → KS=3.530, p=0.532

Carb. PAN/CGO:

Power-law → KS=0.709, p=0.980
 Exponential → KS=inf, p=0.000
 Log-normal → KS=3.540, p=0.526

Nodes vs Average Degree (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Power-law → KS=5.951, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=16.772, p=0.000

Carb. PAN:

Power-law → KS=8.953, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=21.402, p=0.000

PAN:

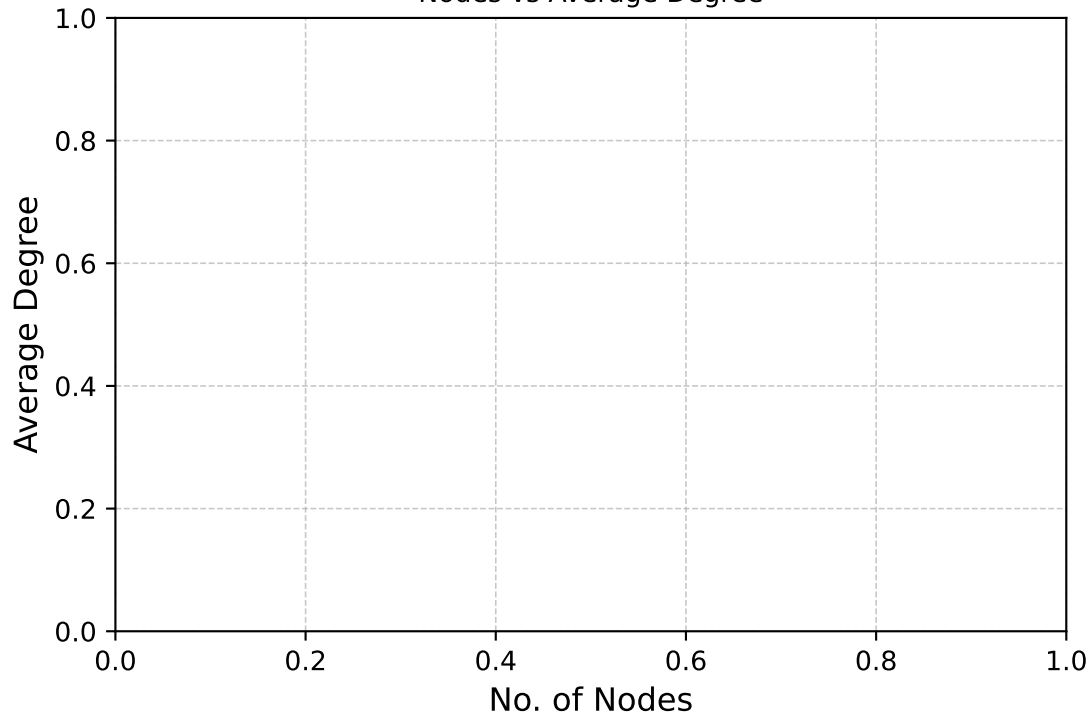
Power-law → KS=5.290, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=17.650, p=0.000

Carb. PAN/CGO:

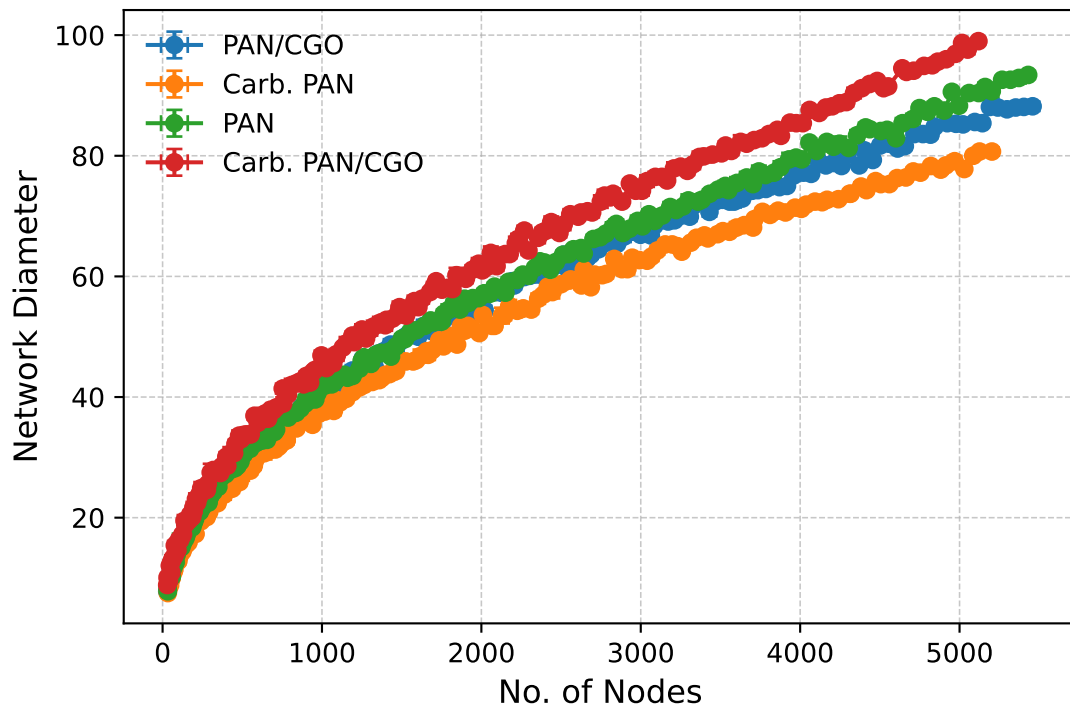
Power-law → KS=4.979, p=0.000
 Exponential → KS=inf, p=0.000
 Log-normal → KS=18.453, p=0.000

LogNormal Fit: $y = a \cdot \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$

Nodes vs Average Degree



Nodes vs Network Diameter (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Power-law → KS=2.850, p=0.314
 Exponential → KS=inf, p=0.000
 Log-normal → KS=2.002, p=0.639

Carb. PAN:

Power-law → KS=3.091, p=0.213
 Exponential → KS=inf, p=0.000
 Log-normal → KS=2.136, p=0.587

PAN:

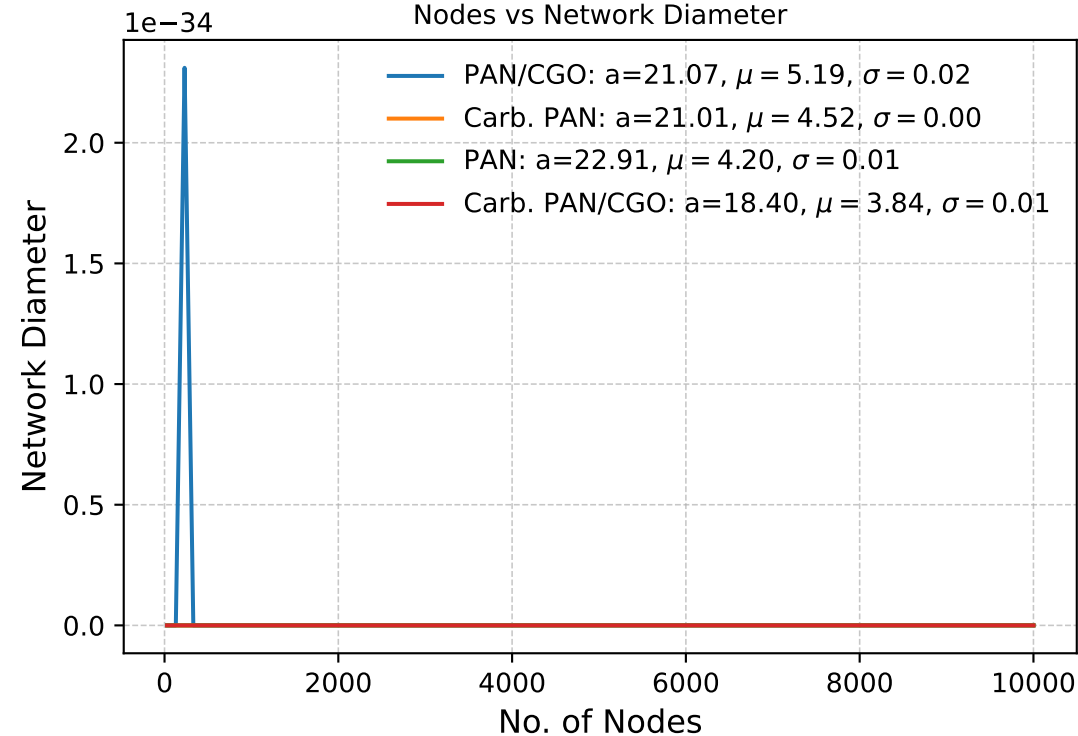
Power-law → KS=2.491, p=0.378
 Exponential → KS=inf, p=0.000
 Log-normal → KS=1.815, p=0.665

Carb. PAN/CGO:

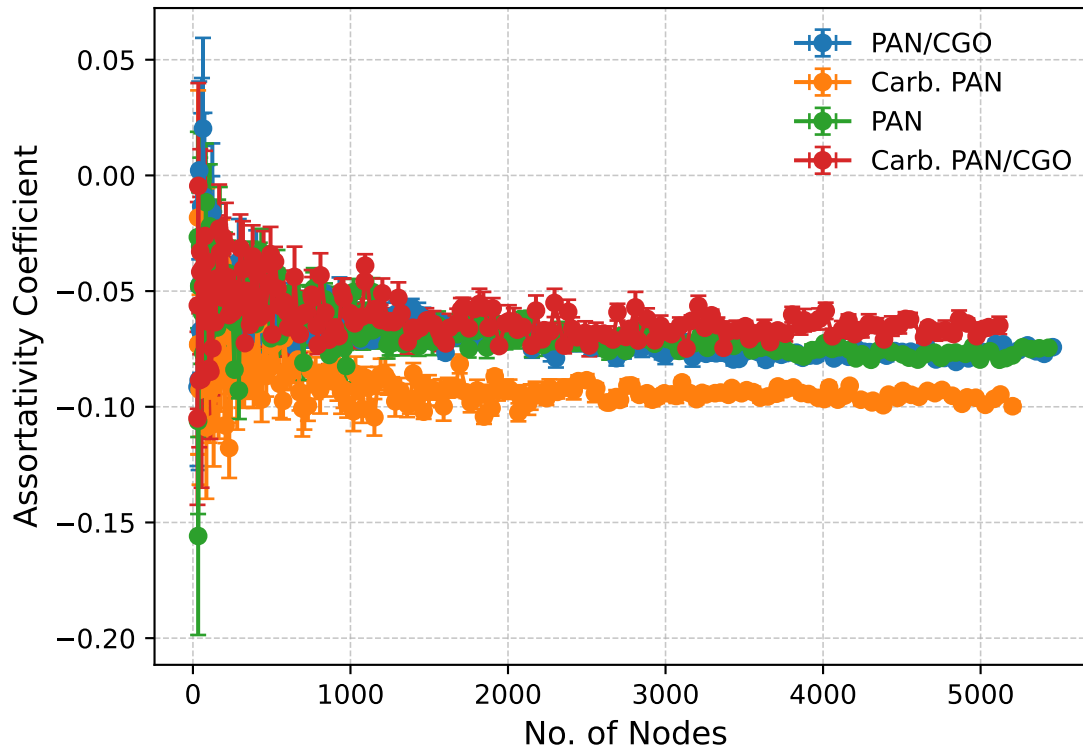
Power-law → KS=2.210, p=0.527
 Exponential → KS=inf, p=0.000
 Log-normal → KS=1.753, p=0.643

LogNormal Fit: $y = a \cdot \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$

Nodes vs Network Diameter



Nodes vs Assortativity Coefficient (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Power-law → KS=35.994, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=3.085, p=0.000

Carb. PAN:

Power-law → KS=38.995, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=51.545, p=0.006

PAN:

Power-law → KS=39.628, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=6.964, p=0.000

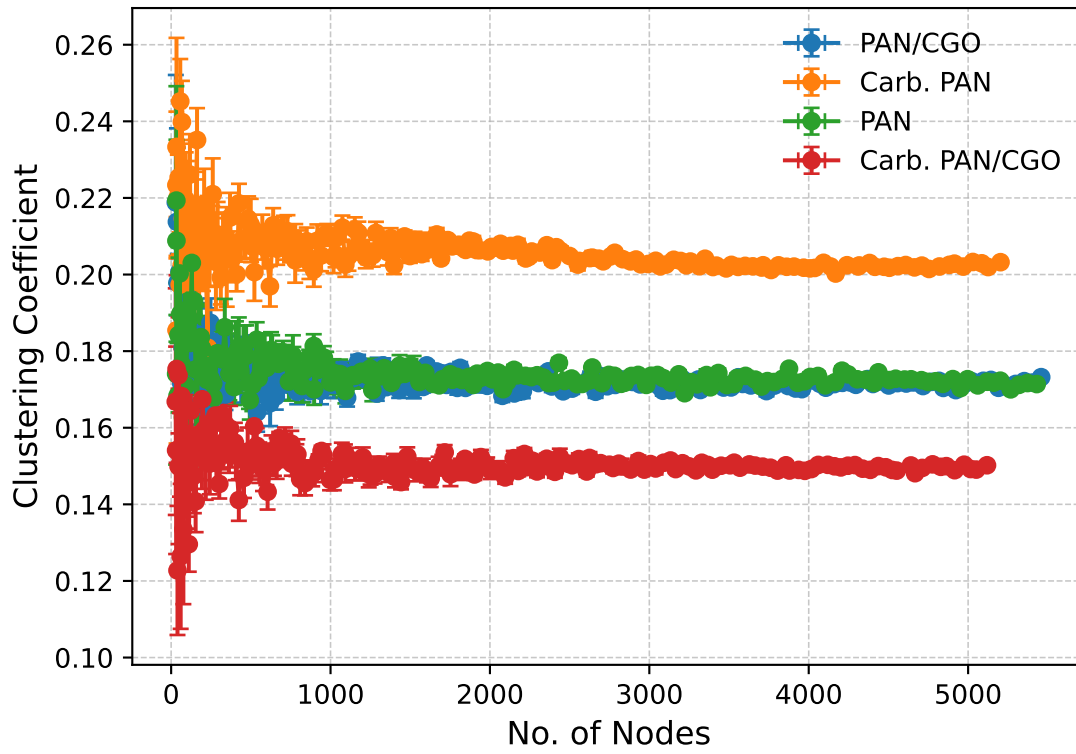
Carb. PAN/CGO:

Power-law → KS=40.866, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=4.540, p=0.000

Nodes vs Clustering Coefficient (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Power-law → KS=42.842, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=8.315, p=0.000

Carb. PAN:

Power-law → KS=43.861, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=8.241, p=0.000

PAN:

Power-law → KS=49.325, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=10.674, p=0.000

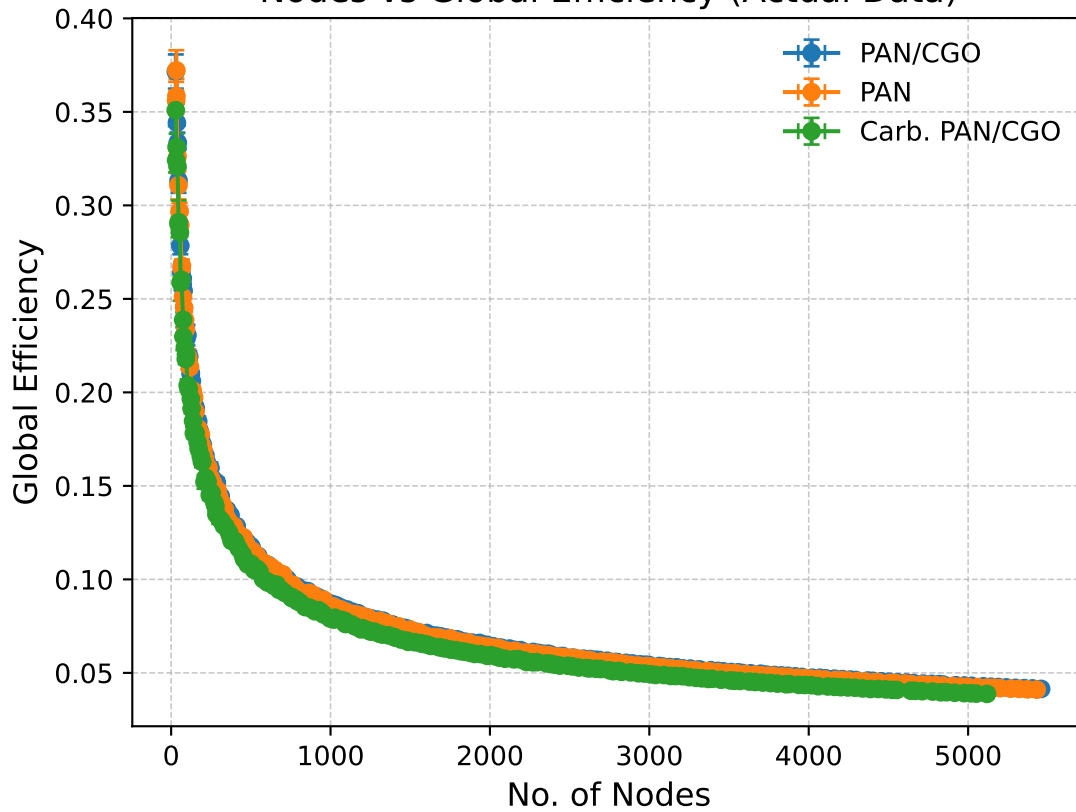
Carb. PAN/CGO:

Power-law → KS=46.188, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=13.125, p=0.000

Nodes vs Global Efficiency (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Power-law → KS=12.147, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=0.478, p=0.177

PAN:

Power-law → KS=12.039, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=0.489, p=0.163

Carb. PAN/CGO:

Power-law → KS=11.506, p=0.000

Exponential → KS=inf, p=0.000

Log-normal → KS=0.507, p=0.148